

outward fluxes of B , C , and A can be obtained through equations 60a, 60b, and 60c, respectively. This model can also be applied to the symport transport mechanism of three substrates in a ratio of $A : B : C$ [100].

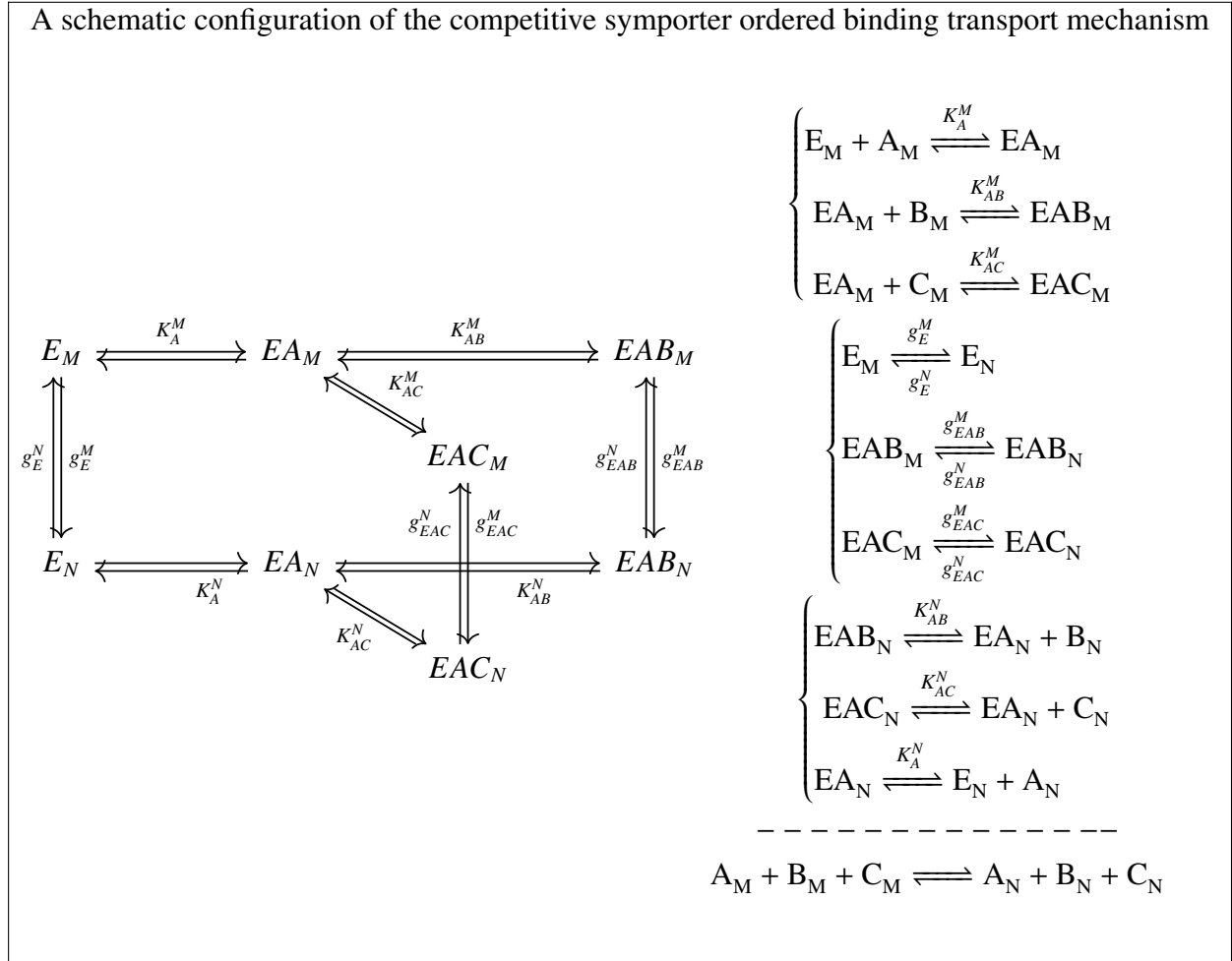


Figure 16: A schematic configuration of the competitive symporter ordered binding transport mechanism of two solutes and the regarding net flux transport equation. In this configuration, the carrier exists in four different states on each surface of the membrane. Transport equations are reparametrized and transformed into the general resistance form from Layton model [100]. Dissociation constants (K_A , K_{AB} , K_{AC} terms), and the translocation rate constants (g_E , g_{EAC} , and g_{EAB} terms) are depicted in the reaction cycle in the right panel. Subscripts M and N, denote the M and N sides of the cell membrane (part 1/2 continued on next page).

Net Transport Flux from M to N

$$J_{B,symporter}^{M,N(net)} = [E]_t \frac{(g_{EAB}^M \alpha^M \beta^M) R_{NN} - (g_{EAB}^N \alpha^N \beta^N) R_{MM}}{R_M R_{NN} + R_N R_{MM}} \quad (60a)$$

$$J_{C,symporter}^{M,N(net)} = [E]_t \frac{(g_{EAC}^M \alpha^M \gamma^M) R_{NN} - (g_{EAC}^N \alpha^N \gamma^N) R_{MM}}{R_M R_{NN} + R_N R_{MM}} \quad (60b)$$

$$J_{A,symporter}^{M,N(net)} = [E]_t \frac{(g_{EAB}^M \alpha^M \beta^M + g_{EAC}^M \alpha^M \gamma^M) R_{NN} - (g_{EAB}^N \alpha^N \beta^N + g_{EAC}^N \alpha^N \gamma^N) R_{MM}}{R_M R_{NN} + R_N R_{MM}} \quad (60c)$$

where

$$[E]_t = [E]_M + [EA]_M + [EAB]_M + [EAC]_M + [EAC]_N + [EAB]_N + [EA]_N + [E]_N$$

$$\alpha^M = \frac{[A]_M}{K_A^M}, \beta^M = \frac{[B]_M}{K_{AB}^M}, \gamma^M = \frac{[C]_M}{K_{AC}^M} \quad \alpha^N = \frac{[A]_N}{K_A^N}, \beta^N = \frac{[B]_N}{K_{AB}^N}, \gamma^N = \frac{[C]_N}{K_{AC}^N}$$

$$R_M = 1 + \alpha^M + \alpha^M \beta^M + \alpha^M \gamma^M \quad R_N = 1 + \alpha^N + \alpha^N \beta^N + \alpha^N \gamma^N$$

$$R_{MM} = g_E^M + g_{EAB}^M \alpha^M \beta^M + g_{EAC}^M \alpha^M \gamma^M \quad R_{NN} = g_E^N + g_{EAB}^N \alpha^N \beta^N + g_{EAC}^N \alpha^N \gamma^N$$

Figure 16: A schematic configuration of the competitive symporter ordered binding transport mechanism of two solutes and the regarding net flux transport equation. In this configuration, the carrier exists in four different states on each side of the membrane. Transport equations are reparametrized and transformed into the general resistance form from Layton model [100]. Dissociation constants (K_A , K_{AB} , K_{AC} terms), and the translocation rate constants (g_E , g_{EAC} , and g_{EAB} terms) are depicted in the reaction cycle in the right panel. Subscripts M and N, denote the M and N sides of the cell membrane. $[E]_t$ is the total number of transporter per unit area of the membrane (part 2/2 continued from previous page).

Symporter simplified model:

To conclude the section on symporters, a commonly used reduced symporter carrier model is presented. In this model, the rapid-equilibrium assumption, quasi-steady-state assumption, and symmetric carrier assumption ($K_A^M = K_A^N = K_A$, $K_B^M = K_B^N = K_B$, $K_C^M = K_C^N = K_C$) are invoked. Furthermore, it is assumed the binding and unbinding steps do not follow a particular sequential order. Under these assumptions, for the general case of a symporter transporting three solutes in the ratio of $aA : bB : cC$, the net outward symporter turnover [rate](#) is given by equation 61a and the corresponding solute fluxes of A, B, and C are given by equations 61b, 61c, and 61d, respectively.